

4.1 Analytical Techniques

Question Paper

Course	CIEA Level Chemistry
Section	4. Analysis
Topic	4.1 Analytical Techniques
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

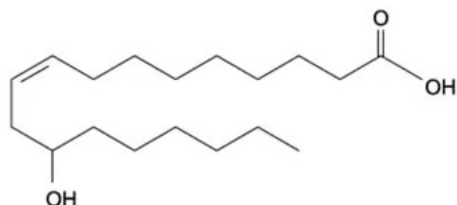
Which of the following corresponds to the base peak in a mass spectrum?

- A. The peak corresponding to the ion with lowest m/e .
- B. The peak corresponding to the molecular ion.
- C. The peak corresponding to any ion formed by the fragmentation of the molecule.
- D. The peak corresponding to the most abundant ion.

[1 mark]

Question 2

The diagram shows the structure of ricinoleic acid.



Which fragment would produce the tallest peak in the mass spectrum of ricinoleic acid at $m/e = 55$?

- A. CH_2CO_2^+
- B. $\text{CH}=\text{CHCHOH}^+$
- C. C_4H_7^+
- D. COOH^+

[1 mark]

Question 3

In a mass spectrometer, the ions are separated.

Which of the following species will be deflected to the greatest extent?

- A. $^{37}\text{Na}^+$
- B. $^{35}\text{Na}^+$
- C. ^{37}Na
- D. $^{35}\text{Na}^{2+}$

[1 mark]

Question 4

Which of the following statements about propanal, $\text{CH}_3\text{CH}_2\text{CHO}$, and propanone, CH_3COCH_3 is **not** correct?

The compounds have:

- A. Molecular ion peaks at different mass to charge ratios.
- B. Different fragmentation patterns in the mass spectrum.
- C. Absorption in the infrared spectrum due to the carbonyl group.
- D. A different fingerprint region in the infrared spectrum.

[1 mark]

Question 5

A sample of silicon contains the following isotopes

Isotope	Percentage abundance
^{28}Si	81.21
^{29}Si	14.10
^{30}Si	4.69

What is the relative atomic mass of silicon in this sample, to 1dp?

- A. 29.8
- B. 29.0
- C. 28.2
- D. 28.0

[1 mark]

Question 6

Chlorine exists as two isotopes with mass numbers 35 and 37. A sample of chlorine contains 75.3% of chlorine-35 atoms.

What is the relative atomic mass of chlorine in this sample, to 2dp?

- A. 35.16
- B. 36.00
- C. 35.49
- D. 35.00

[1 mark]**Question 7**

A sample of neon contains the following isotopes

Isotope	Percentage abundance
^{20}Ne	90.92
^{21}Ne	0.26
^{22}Ne	8.82

What is the relative atomic mass of neon in this sample, to 2dp?

- A. 21.00
- B. 20.18
- C. 20.09
- D. 20.00

[1 mark]

Question 8

Nitrogen trichloride has the formula NCl_3 , a sample was found to contain only nitrogen atoms with mass number 14, and chlorine atoms with mass numbers 35 and 37.

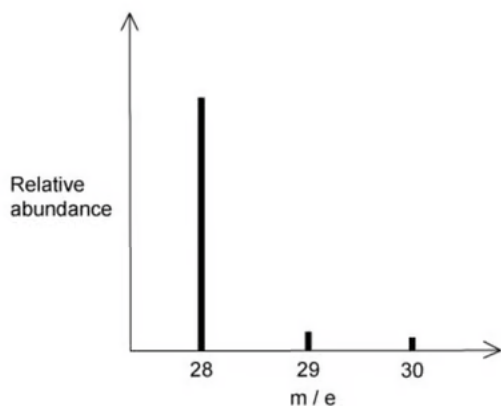
Which of the following ions and mass/charge ratio for the ions responsible for the molecular ion peaks in the mass spectrum of nitrogen trichloride is **incorrect**?

	Ion	m/e
A	$\text{N}({}^{35}\text{Cl})_3^+$	119
B	$\text{N}({}^{35}\text{Cl})_2{}^{37}\text{Cl}^+$	121
C	$\text{N}({}^{35}\text{Cl})({}^{37}\text{Cl})_2^+$	123
D	$\text{N}({}^{37}\text{Cl})_2^+$	125

[1 mark]

Question 9

The mass spectrum of silicon is shown below.



What is the best estimate for the relative atomic mass of silicon in this sample?

- A. 29.0
- B. 28.8
- C. 28.3
- D. 28.0

[1 mark]

Question 10

Which of the following cannot be obtained from an infrared spectrum?

- A. The molecular mass
- B. The presence of C=O bonds
- C. The presence of O-H bonds
- D. The identity of a compound through comparison with other spectra

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- Mass spectrometry can be used to determine the molecular formula of a compound from the mass of the molecular ion.
- The peak representing the **base peak** is the most intense (tallest) peak in the **mass spectrum** due to the ion with the greatest molecular abundance.

A is incorrect as the base peak is not formed by the lowest m/e value ion.

B is incorrect as the molecular ion peak does not necessarily correspond to the base peak. The peak representing the **molecular ion (M^+ or $M_{\bullet}^+$)** is called the **molecular ion peak**. Excluding any peaks due to the presence of heavier isotopes, this is the peak with the highest m/e ratio.

C is incorrect as the base peak is not formed by any ion that is fragmented from the molecule.

2

The correct answer is **C** because:

- The question has asked for the tallest peak at 55; this will be the fragment that is most abundant at that m/e value.
- $[C_4H_7]^+$ has an m/e value of 55.

A is incorrect as the fragment $CH_2CO_2^+$ would have a peak at 58.

B is incorrect as the fragment $CH=CHCHOH^+$ would have a peak at 56.

D is incorrect as the fragment $COOH^+$ would have a peak at 45.

3

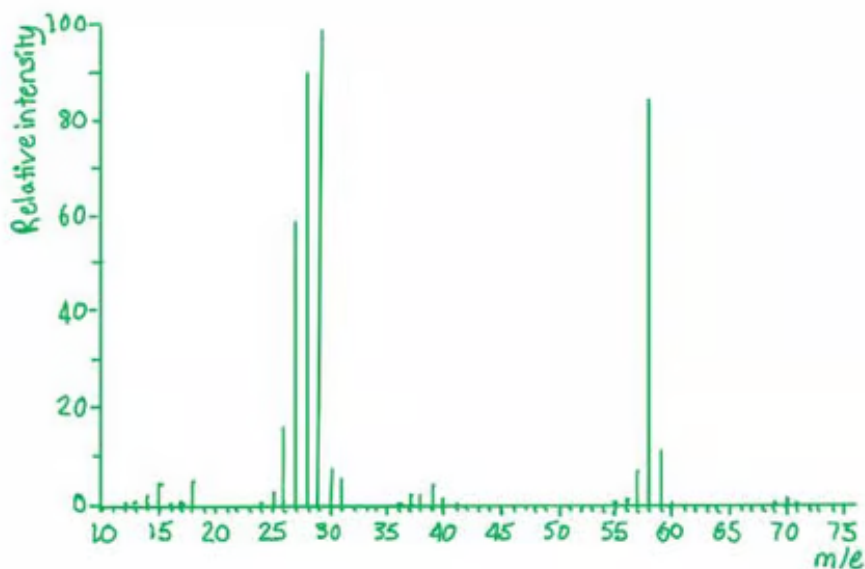
The correct answer is **D** because:

- The ion that will be deflected the most in the mass spectrometer is the one with the smallest m/e value.
- They will have the largest positive charge and will be more attracted to the negative pole of the magnet within the mass spectrometer so have a larger deflection.

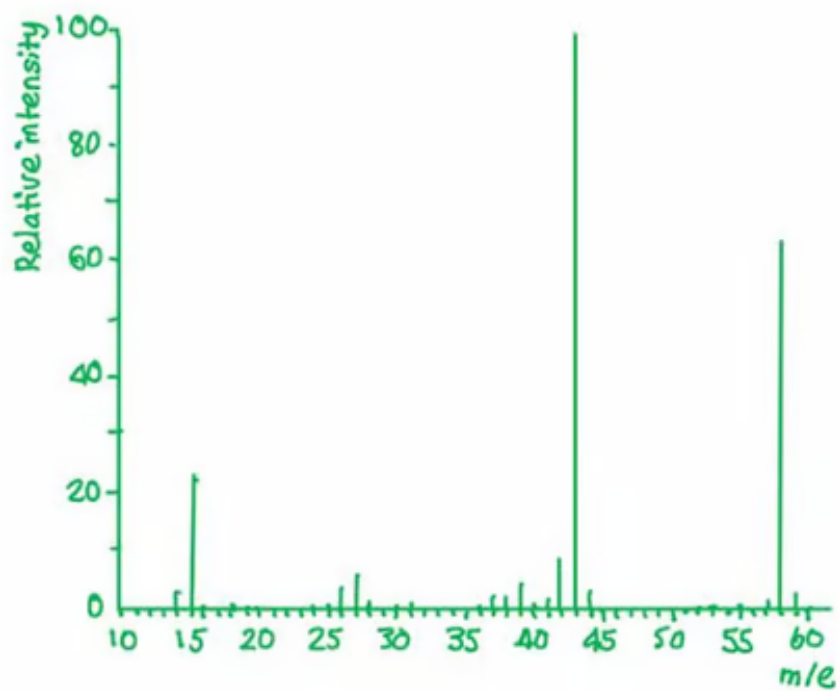
4

The correct answer is **A** because:

- The peak representing the **molecular ion (M^+ or $M_{\bullet}^+$)** is called the **molecular ion peak**. Excluding any peaks due to the presence of heavier isotopes, this is the peak with the highest m/e ratio.
- Both propanal and propanone have a molecular ion peak at $m/e = 58$
 - Propanal arises from the $\text{CH}_3\text{CH}_2\text{CHO}^+$ ion.
 - Propanone arises from the $\text{CH}_3\text{COCH}_3^+$ ion.
- Propanal mass spectrum



- Propanone mass spectrum





B is incorrect as they would have different patterns in the mass spectrum.

C is incorrect as they would both have absorption in the infrared spectrum as both molecules have a C=O bond.

D is incorrect as they would have a different fingerprint region in the infrared spectrum.

5

The correct answer is **C** because:

- **To calculate the relative atomic mass of silicon**
 - Multiply each isotopic mass by its percentage abundance
 - Add the figures together
 - Divide by 100

- **A_r of silicon**

$$\circ A_r = \frac{(28 \times 81.21) + (29 \times 14.10) + (30 \times 4.69)}{100} = 28.2348$$

A is incorrect as the percentages have been mixed up.

B is incorrect as this is the average mass without taking into account the abundances.

D is incorrect as this is the most abundant isotope.

6

The correct answer is **C** because:

- **To calculate the relative atomic mass of chlorine**
 - Multiply each isotopic mass by its percentage abundance
 - Add the figures together
 - Divide by 100

- **A_r of Chlorine**

$$\circ A_r = \frac{(35 \times 75.3) + (37 \times 24.7)}{100} = 35.494$$

A is incorrect as this answer has used 36 as the other isotope mass.

B is incorrect as this is the average without taking into account the abundance.

D is incorrect as this is the most abundant mass.

7

The correct answer is **B** because:

- **To calculate the relative atomic mass of neon**
 - Multiply each isotopic mass by its percentage abundance
 - Add the figures together
 - Divide by 100
- **A_r of neon**
 - $A_r = \frac{(20 \times 90.92) + (21 \times 0.26) + (22 \times 8.82)}{100} = 20.179$

A is incorrect as this is the average without taking into account the abundance.

C is incorrect as in this answer, the percentages are mixed up.

D is incorrect as this is the most abundant mass.

8

The correct answer is **D** because:

- The question states that there are only nitrogen atoms with a mass of 14.
- There are two isotopes of chlorine ^{35}Cl and ^{37}Cl ; this will create ions with the following combinations:
 - All 3 chlorine atoms made up of ^{35}Cl
 - 2 ^{35}Cl and 1 ^{37}Cl
 - 2 ^{37}Cl and 1 ^{35}Cl
 - All three chlorine atoms made up of ^{37}Cl
- Therefore, ion **D** should be $\text{N}(^{37}\text{Cl})_3^+$

A, B & C are incorrect as they are all correct ions and m/e values.

9

The correct answer is **C** because:

- Reading the relative abundances from the graph:
 - 28 – 80%
 - 29 – 12%
 - 30 – 8%
- **To calculate the relative atomic mass of silicon**
 - Multiply each isotopic mass by its percentage abundance
 - Add the figures together
 - Divide by 100
- **A_r of silicon**
 - $A_r = \frac{(28 \times 80) + (29 \times 12) + (30 \times 8)}{100} = 28.28$

10

The correct answer is **A** because:

- Infrared is the analysis of infrared light interacting with a molecule; this can be measured by absorption, emission and reflection.
- It is used to determine functional groups in atoms based on the bond strength and mass of the atoms.
- It is not used to determine the molecular mass; a mass spectrometer does this.

B, C & D are incorrect as these are all possible information that can be obtained by the infrared spectrum of a molecule.

